Acquiring drive train insights by using the electric motor as a lateral and torsional vibration sensor

Using voltage-extended MCSA for fault detection and root cause analysis throughout industrial drive trains

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Technical white paper courtesy of

Samotics

Abstract

The implementation of condition-based maintenance strategies in current industrial drive trains is a major contributor to the increasing uptime in electric machinery, resulting in significant economical gains. Condition monitoring techniques are generally based on interpreting the physical fingerprints of faulty machines. Although the measurement of mechanical vibrations is very popular within this field of expertise, there are several supplementary benefits to using motor current signature analysis (MCSA). This technique uses the electric motor as a sensor: a safe, cost-effective and robust solution. **By interpreting the motor currents as mechanical vibrations, it is often not only possible to detect developing faults, but also to link them to the root cause, enabling proactive action.** This paper provides an in-depth look at the electromechanical foundations that underpin MCSA, including the algorithms that enable drive-train-wide analysis, and concludes with some striking industrial examples.

About the authors

Bram Vervisch and Bram Corne earned their doctorates in electromechanical engineering at Ghent University. In January 2018 they founded ORBITS, a company that assists clients in optimizing their dynamic rotating equipment from design through operation. “The two Brams” provide analytical and finite element modeling, troubleshooting and root cause failure analysis, and coaching on signal processing, data handling and electrical/mechanical systems and processes. Both men are skilled ISO 18436 Cat. III and Cat. IV vibration specialists and MCSA experts. **orbits.be**

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1 Introduction

Only a century ago, steam and fuel engines were shaping the industrial landscape. Today, electric machines dominate, in both abundance and excellence. Industrial electrification enabled the efficient and instantaneous transportation of massive amounts of energy throughout each and every plant. Unlike traditional steam and combustion engines, the electric machine can work in both directions: as a generator, to turn mechanical power into electrical energy, and as a motor to turn alternating current into rotating mechanical work.

Although electric machines are extremely robust and reliable, unexpected failures still regularly occur. In the last decade, European industry suffered an average downtime of 6.3 percent due to failing electric machines [1]. This loss of production can be translated into a roughly estimated economic loss of one trillion euros. As a consequence, intensive industrial and academic efforts have fueled the development of effective maintenance strategies. By correctly maintaining rotating equipment, sudden breakdowns can be reduced by roughly 50 percent [1].

Condition-based maintenance

One of the more advanced maintenance strategies is referred to as condition-based maintenance, or predictive maintenance. In essence, attempts are made to estimate the actual, or even future, condition of the machine in order to plan and schedule proper maintenance actions before the machine breaks down. For example, the detection of a developing bearing failure enables the asset owner to order parts and prepare the overhaul at a convenient time before the drive train suddenly collapses, without unnecessary actions while the bearing is still sufficient.

The bottleneck in this approach is the exact and reliable pinpointing of all developing problems in the scoped drive train. Not only false negatives (the inability to detect a problem) but also false positives (reporting a problem which is not present) can

easily undo the effectiveness of the condition-based maintenance strategy [2]. This strategy therefore stands or falls on the accuracy of the system used to extract the healthy/faulty fingerprint of the machine under scope.

There are several technologies that can complete this crucial task. Generally, any parameter or signal that is able to indicate aberrant machine behavior can be used to monitor the machine's condition [3]. Inherent diagnostic parameters are typically physical quantities such as temperature, sound, current, vibration, flux and so forth.

Complexity $\neq$ effectiveness

Each of these signals can be implemented on several levels of sophistication. For example, a stator winding thermocouple reveals less diagnostic information than an infrared thermographic analysis. Similarly, an RMS velocity vibration sensor does not provide the same detailed data as a spectral vibration analysis. But beware: the level of complexity does not dictate the effectiveness of the condition monitoring system. Its effectiveness is determined by two parameters. The first is the failure mode probability: how likely is that component of the machine to fail? The second is the failure mode propagation: how easily can the monitoring system detect failure-related changes before breakdown?

Focusing on the electric motor itself, it has been shown that monitoring systems based on vibration and on current are the most effective (and cost-effective) technologies to spot the most frequent failure modes before they induce unexpected downtime or significant collateral drive train damage [4]. The aim of this paper is to show how the basic motor current signature analysis (MCSA) approach can be extended to enable equally effective insights into the entire drive train of a rotating industrial asset.

We will start by answering the above questions on probability and propagation for faults within the electric motor, then see how we can expand the scope of MCSA diagnosis to include the transmission and load. We will conclude with four striking

industrial examples.

But first, why might one want to analyze electrical signals instead of vibration, temperature, sound and so forth in the first place?

2 The potential of voltage-extended MCSA

Analyzing electrical signals instead of any other physical quantity enables us to measure in a clean and safe environment. Because currents remain equal throughout the motor feeder line, it does not matter where the current measurements are conducted [5]. Consequently, the MCSA-based condition monitoring system does not suffer from the industrial environment and its well-known challenges. Machines located in a rough, dusty, ATEX, inaccessible, submersible or otherwise inhospitable environment are equally and easily monitored from the machine control room. This also means there are no challenges regarding IT connectivity and remote control for online and cloud-based monitoring systems.

MCSA is the only technique that allows a thorough evaluation of both electrical and mechanical systems. For the most part, issues arising in the grid, a variable frequency drive (VFD), an electric motor and a mechanical load reflect unique signatures in the stator current signals [6]. Additionally, in contrast with vibration data captured by accelerometers, problems are revealed before they actually result in physical damage. For example, indirect causes of bearing failure can be detected and evaluated before raceway bearing damage (a direct cause of failure) is even present. MCSA uses the electric machine as a sensor and enables direct diagnostics from the beating heart of the drive train.

Capture the asset's exact operating state

Extending the basic MCSA approach to include voltage signals adds another techni-

cal advantage: the captured electrical data contains the exact operating state conditions of the drive train [7]. Using current and voltage signals, exact speed and torque quantities can be extracted. As a result, the presence of potential fault-indicating components can be related to the operating state. This enables the analyst to validate and differentiate between several potential and suspected failure modes. For example, the analyst can determine whether a speed-dependent frequency component is being caused by an unbalance or a misalignment by interpreting the speed dependency of the current fault pattern [8]. Other monitoring technologies may include the operational speed parameter, but are rarely able to obtain a reliable torque estimation [9]. It is, for example, very challenging to evaluate the gear mesh quality by comparing two vibration signals at unknown torque conditions (even at the same speed). The torque has a significant effect on the magnitude of gear-related components in these spectra [10]. As MCSA includes the exact operational load condition, pragmatic and reliable conclusions can be drawn even when the operational conditions were not equal.

We now turn to the details.

3 From fault to detection

A failure is defined as the moment when a machine cannot perform the specific task for which it was designed: for example, a motor-pump group that is no longer able to produce the desired pressure or flow rate. The aim of condition monitoring is to diagnostically interpret changes to a machine's physical signature, in order to signal upcoming failure well before it manifests. This requires that the change imposed by the developing fault generates a signal which propagates from the damaged component toward the monitoring sensor (figure 1). Both this transfer path and the measurement accuracy are of utmost importance.

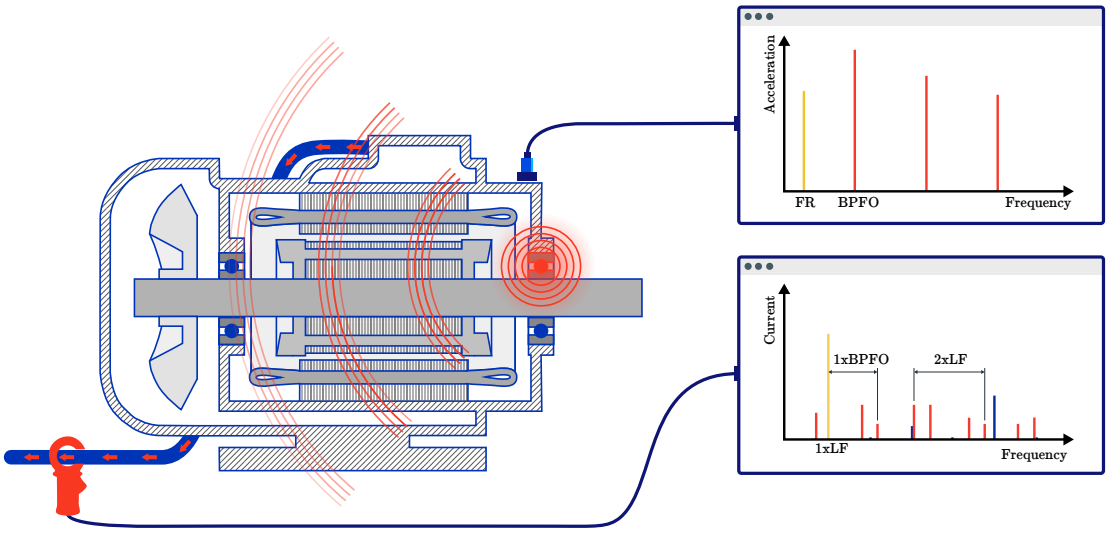

Figure 1. A faulty bearing propagates signals to both a vibration sensor and a current sensor.

3.1 Fault probability in electric motors

An electric motor is basically constructed by assembling three main types of components in one housing: a stator, a rotor and bearings. The most frequent failures therefore originate from one of these three categories. Table 1 summarizes the probability distribution for failure causes in induction motors; more specifically, in low-voltage asynchronous squirrel cage induction motors with distributed windings [11].

	Stator	Bearing	Rotor	Other
> 11 kW	12%	42%	8%	38%
> 75 kW	36%	41%	9%	14%
> 150 kW	37%	41%	10%	12%
> 500 kW	35%	21%	44%	0%

Table 1. Cause-of-failure distribution for low-voltage induction motors [11].

The distribution in table 1 does not entirely give us what we are looking for, however. Generally, there is a distinction between the cause of the shutdown and the actual root cause of the failure. For example, a failed bearing can be the direct reason why an electric motor stops working. But the bearing failure could be caused by degradation of the stator winding, which in turn is caused by a power quality issue—the real reason the failure occurred. The whole failure pattern and its development is referred to as the failure mode. For smaller motors, less effort is usually taken to identify the actual root cause of failure, and thus table 1 does not tell us the whole failure mode story. A more useful breakdown is the one in table 2 [11].

	Stator	Bearing	Rotor	Other
Direct	7%	85%	3%	5%
Indirect	38%	40%	12%	10%

Table 2. Indirect vs. direct cause-of-failure distribution for low-voltage induction motors [11].

For both direct and indirect causes of failure, there is a distinction between mechanical and electrical origin. Most of the time, this separation is easily made and provides insights into both failure mode and fault propagation. Every component or sub-component can function as either a direct or indirect cause of failure [3]. Table 3 summarizes the previously discussed categories by assigning some intuitive examples to their characteristic cause of failure [11].

		Stator	**Bearing**	**Rotor**	**Other**
Direct	Mechanical	Cracked housing; stator looseness	Broken cage; raceway flaking/spalling; grease contamination	Broken shaft coupling; obstructed cooling fan; stator/rotor collision	Environmental influences; excessive dust/moisture; foundation cracks
	Electrical	Phase-to-phase short circuit; loose terminal connection	Large electrostatic discharge; common mode shaft leaking currents	Multiple rotor bar cracks; rotor/stator shaft discharges	Loose connections; electric grid failure
Indirect	Mechanical	Cracked lamination steel; loose end windings, loose slot wedges	Micro pitting; bearing creep; cocked bearing	Misalignment; mechanical unbalance	Machine mounting; poor dimensioning; structural flexibilities/looseness
	Electrical	Inter-turn short circuits; turn-to-turn insulation degradation	Beearing currents; small static discharges; electro pitting	Broken rotor bar; rotor porosity	Power quality issues; long-term grid disturbances

Table 3. Examples of indirect vs. direct, mechanical vs. electrical causes of failure for low-voltage induction motors [11].

Combining tables 2 and 3 gives us the most frequent failure modes for electric motors: bearing failures (direct and indirect), inter-turn short circuits (indirect) and broken rotor bars (indirect). To be effective, a condition monitoring system must be able to detect these modes prior to catastrophic failure. If the diagnostic process also enables an exact characterization of the root cause, then the condition monitoring system will enable you to go beyond predictive maintenance into proactive maintenance, where you can correct the source of the problem (such as a power quality issue) to prevent future recurrence of the whole fault development chain (stator winding degradation and subsequent bearing failure).

3.2 Electrical fault propagation in electric motors

Fault propagation is generally quite straightforward within a given physical area. A mechanical fault leads to a mechanical signature and an electrical fault leads to an electrical signature. In electric machines, however, the two quantities are very closely linked. Mechanical rotor vibrations impose a varying magnetic flux in the motor, producing current variations, and vice versa. The diagnostic interpretation of this relationship requires extensive signal processing techniques and expertise.

Indirect electrical failure modes in the stator commonly cause a deviation in the machine's impedance [9]. By monitoring current signals, these variations can be directly detected, enabling electrical fault detection even though it is not yet severe enough to produce mechanical effects (vibrations, changes in temperature). The shift in impedance distorts the resulting magnetic field distribution; as a result, stator problems can not only be identified through impedance variation, but also through newly introduced spectral components [11]. This approach is very effective in detecting premature stator winding turn-to-turn short circuits.

The analysis of electromagnetic field distortions is also useful to detect broken rotor bars. Similarly to stator winding short circuits, broken rotor bars cause a shift in rotor impedance, resulting in a distorted magnetic field [12]. The frequency content

of the monitored current signal in this case differs from the stator problem case, enhancing the distinction.

Although it is quite easy and pragmatic to determine the exact origin of indirect electrical failure modes, estimating their severity is more challenging. It requires thorough knowledge of the machine's properties and environment, advanced data processing, historical monitoring data, machine failure history and a great deal of expertise to translate the diagnostic results into accurate maintenance advice.

In some cases, electrical rotor and/or stator problems can be observed via comparable vibration patterns, though generally at a later stage of development. These patterns are also highly dependent on the vibration transfer path and the operational conditions, making it even harder to estimate fault severity and come up with the proper corrective advice [13].

3.3 Mechanical fault propagation in electric motors

The propagation of mechanical problems toward the stator currents occurs mainly through vibrations of the rotor with respect to the stator [14]. For example, if one of the motor's bearings is suffering from a raceway deformation, the rotor will move with respect to the stator housing [11]. This movement inevitably results in an airgap variation, introducing a variation in reluctance and so a change in stator inductance. This in turn induces characteristic spectral components in the stator currents [12], enabling early fault detection and diagnosis.

3.4 Moving beyond the electric motor

Unfortunately, only mechanical problems imposing a radial rotor movement with respect to the stator are found in the motor current. This is sufficient for mechanical problems arising within the motor itself, but when mechanical problems arise in rotating equipment flexibly coupled to the electric machine, no fault components

are introduced into the current signals. As a result, traditional MCSA (limited to spectral current analysis) can only evaluate the electric motor, excluding all other drive train components [13].

To overcome this major drawback, current signal acquisition is supplemented with voltage signals. The combination of voltage and current enables the evaluation of instantaneous electromagnetic torque acting between the rotor and the stator [6]. This torque signal can easily reveal mechanical deficiencies beyond the coupling and makes it possible to monitor the entire drive train of rotating equipment. It is important to note that this way of interpreting and analyzing the electrical data requires advanced algorithms which exceed the basic MCSA approach. We turn now to one such algorithm.

4 Interpreting currents as vibrations

Currents and voltages contain a great deal of information, enabling a deep and thorough analysis of the electric motor and its drive train. In essence, they form the exact impedance function in a three-phased representation. The most simplified voltage equation in a single-phased electric machine can be expressed as:

$$v(t) = e(t) + R \cdot i(t) - \frac{\mathrm{d}L(t) \cdot i(t)}{\mathrm{d}t}$$

with R the stator resistance, $i(t)$ the stator current, $e(t)$ the induced electromagnetic force and $v(t)$ the grid voltage connected to the electric machine [11]. The inductance, which is in fact time-dependent, is related to the airgap $a(t)$ by the following simplified equation:

$$L(t) = \frac{\mu_0 \cdot n^2}{\frac{l_{stat}}{\mu_{stat} \cdot A_{stat}} + \frac{a(t)}{A_{air}} + \frac{l_{rot}}{\mu_{rot} \cdot A_{rot}}}$$

In this equation, visually presented in figure 2, all variables are related to the machine's geometry. The inductance will thus start to vary as soon as a geometric change is imposed on the machine [11]. That change might be a displacement of the rotor with respect to the stator. Furthermore, if that rotor movement is specifically characteristic in terms of frequency content, for example due to an outer race bearing fault, the variation in inductance will be accordingly characteristic. As a result, following the simplified voltage equation, these frequency components are found as sidebands around the main fundamental line current (amplitude modulation) [15].

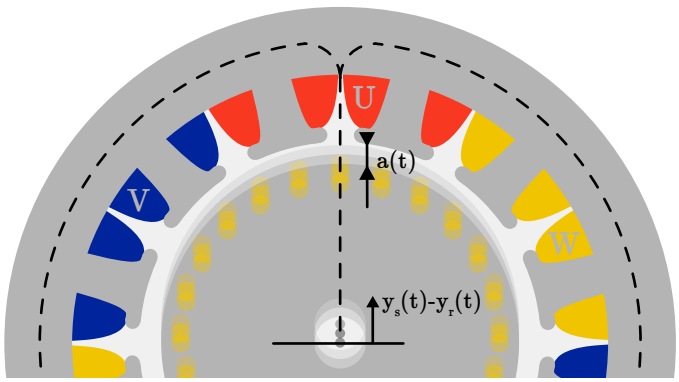

Figure 2. Any geometrical change leads to an inductance variation and current modulation.

To obtain only the fault-related variation in inductance, the current signal must be demodulated. Applying the extended Park's vector approach (EPVA), the three-phased line currents are translated to a two-dimensional coordinate [12] in a reference frame that rotates at the fundamental frequency. This results in a demodulated and simplified current signal as a measure for varying inductance. Representing the inductance as a single row signal significantly simplifies the data interpretation. In essence, the EPVA virtually transforms the electric motor into a proximity probe.

The EPVA represents the lateral rotor vibrations relative to the stator housing and enables effective mechanical diagnostics. Drive train problems that introduce a specific and preferably unique rotor movement relative to the stator are uniformly reflected in the EPVA [12].

With the addition of the voltage signal, an accurate estimation of instantaneous torque can also be obtained from the electrical data. The fast Fourier transform (FFT) is used to transform the time-based current and voltage waveforms into the frequency domain, where they can be used to match aberrant components with the kinematic model of the drive train. As with regular vibration analysis, the frequency components can be linked directly to the kinematics, resulting in a fact-based rather than theoretical diagnosis.

This enables the analyst to diagnose each and every component that generates a possibly fault-related torque ripple [7]. Most problems in pumps, fans, compressors, gears, belts, couplings, etc. are reflected in this signal, with one exception. Although there is a chance, it is unlikely that problems in bearings located further down the drive train will propagate through a detectable current-based torque and/or lateral vibration ripple.

Now let us look at four industrial cases to make our discussion more tangible.

5 Case studies

The electrical data analyzed in each of these industrial examples was captured by the SAM4 condition monitoring system made by Samotics. SAM4's data acquisition system captures 3 x 24-bit voltage and 3 x 24-bit current signals at 20 kS/s/ch. The time waveforms are synchronously captured and transformed into frequency signals with additional signal conditioning and processing techniques. SAM4's anomaly detection algorithms evaluate the periodically captured data and auto-diagnose aber-

rant conditions, which are then further analyzed by the company's data science team and a team of external specialist consultants (including us). The four cases below were intentionally chosen to elucidate both the strengths and weaknesses of MCSA.

5.1 Coupling failure

A 7.5 kW squirrel cage induction machine is driving a hot strip mill application in a steel production plant. The machine speed is 975 rpm at a nominal feeder line frequency of 50 Hz. The machine is fed by a variable frequency drive setup with a fixed rotational speed of 183 rpm, corresponding to a line frequency of 9.75 Hz. The automated anomaly detection algorithm identifies an aberrant pattern evolution in the processed frequency spectrum. A closer look at the data in the mechanical spectrum (EPVA signal) reveals an increasing trend in the spectral components related to the rotational speed, as depicted in figures 3 and 4.

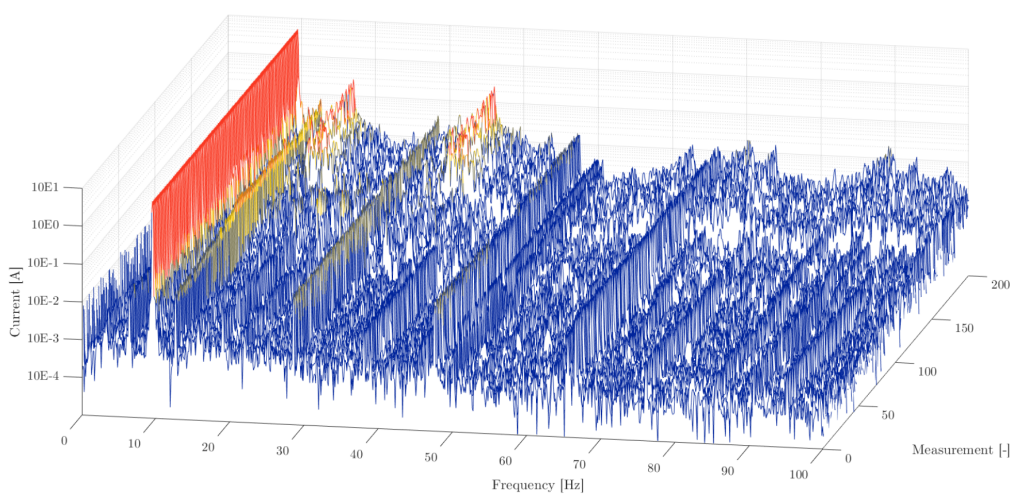

Figure 3. Waterfall plot of the line current spectrum.

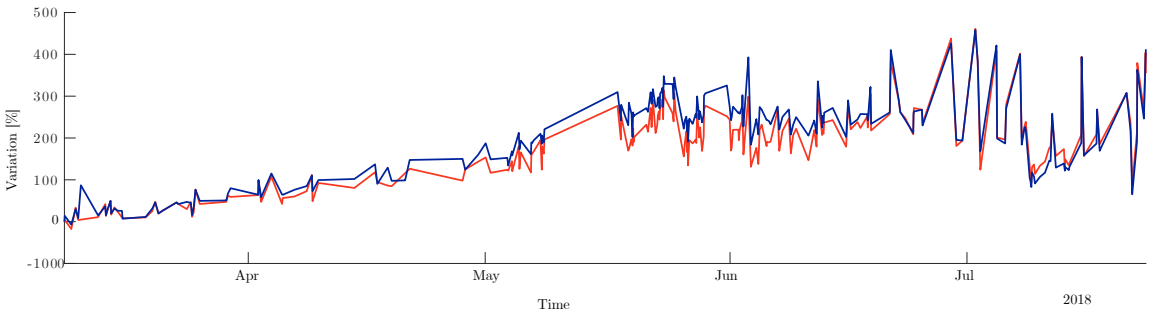

Figure 4. The mechanical spectrum with rotational speed components before (blue) and after (red) the coupling broke.

This trend indicates the presence of evolving drive train looseness or clearance. At a specific moment in time, the prominent fault components transform into a raised noise floor. The fact that periodic fault frequencies are changing toward non-periodic impacts is highly indicative of a a cracked or broken component (system non-linearity).

Considering the main fault mode probabilities in this drive train, SAM4's data science team diagnosed a broken coupling. The diagnosis was validated by the asset owner, who found that one tooth of the coupling had broken and was able to act before further damage caused sudden breakdown.

The electrical data uniformly indicated the presence of a mechanical drive train problem before machine failure occurred. By taking this system's failure modes and their probabilities into account, the data science team was able to provide precise advice to the mill's maintenance team.

5.2 Gearbox clearance

Gases are transported and compressed into a chemical reactor vessel by a 200 kW

two-pole-pair induction machine. Between the electric motor and the compressor, a gearbox with a ratio of 26/119 (expressed in gear teeth) is installed to reduce the operational speed. The machine operates under various load and speed conditions, fully automated by the demands of the chemical process.

The machine operator suddenly hears a significant increase in noise coming from the gearbox. To find the origin of the noise and identify potential failure modes, the torsional spectrum is obtained from the electrical data and thoroughly analyzed. As shown in figure 5, comparing the torsional spectrum at two moments in time reveals that the gear meshing frequency and its sidebands are slightly increasing in magnitude. This is a solid indication that some radial gear clearance has been introduced between the two meshing gears.

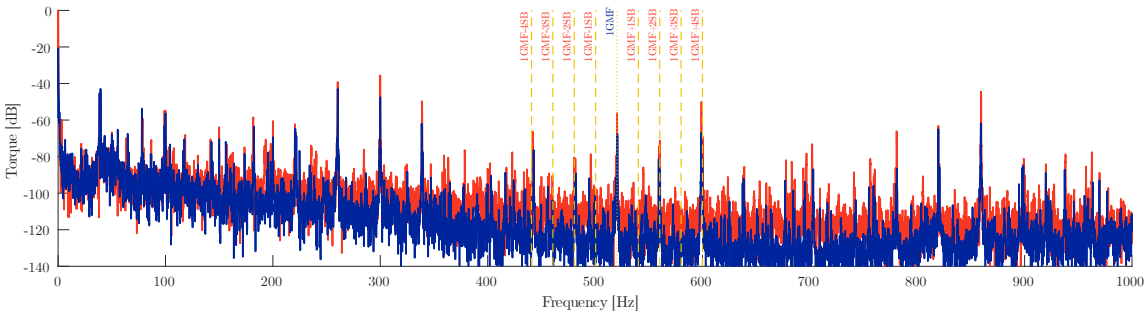

Figure 5. The torsional spectrum with gear meshing components before (blue) and after (red) additional clearance.

As data from several operating points is captured and included in the analysis, the speed dependency of the gear mesh frequency is then evaluated to gain additional insight into the problem. Figure 6 indicates that the aberrant clearance is highly speed dependent. Moreover, at around 90 percent of the nominal speed, the gear mesh frequency components are significantly larger than at both higher and lower speeds. The most reasonable conclusion is that the meshing clearance is enabling

the excitation of the input shaft's torsional natural frequency. That explains why the gearbox is generating a specific noise under very specific operational conditions.

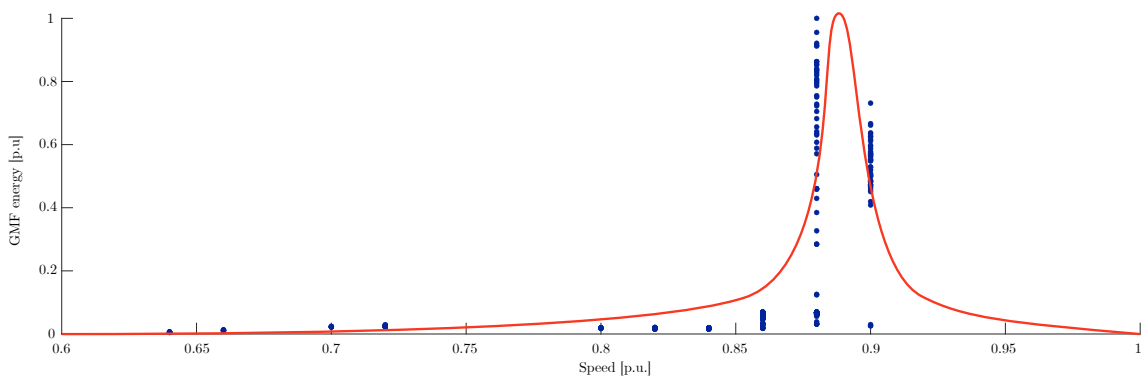

Figure 6. Meshing frequency as a function of the operational input shaft speed (blue) with fitting curve (red).

This diagnostic analysis led to overhauling the gearbox, which resolved the radial clearance between the gears and enhanced the overall meshing quality. All aberrant noise and torsional vibrations disappeared.

By combining accurate operating state information and electrically obtained torque signals, the data science team gained insight into the proper working of system components further down the drive train. These insights enabled the team to advise the plant's maintenance department as to the most adequate corrective action, a gearbox overhaul.

5.3 Screw deformation

In a water treatment facility, sludge is transported between two vessels using an Archimedes' screw. An 11 kW two-pole induction motor drives that slowly rotating

shaft via a planetary gearbox. The anomaly detection algorithm detects an increase in several spectral components that are directly related to the position of the rotor within the stator housing. The mechanical lateral vibration signal, depicted in figure 7, shows a clear rotor deformation. Components related to the rotational speed of the motor and the rotor bar passing frequency with sidebands are vastly increasing, as can be observed in the trend presented in figure 8.

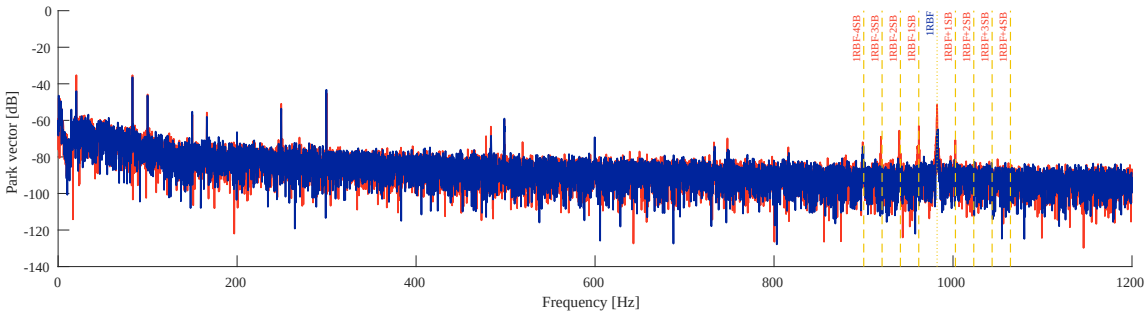

Figure 7. Two snapshots of the demodulated current frequency spectrum with (red) and without (blue) rotor deformation frequency components.

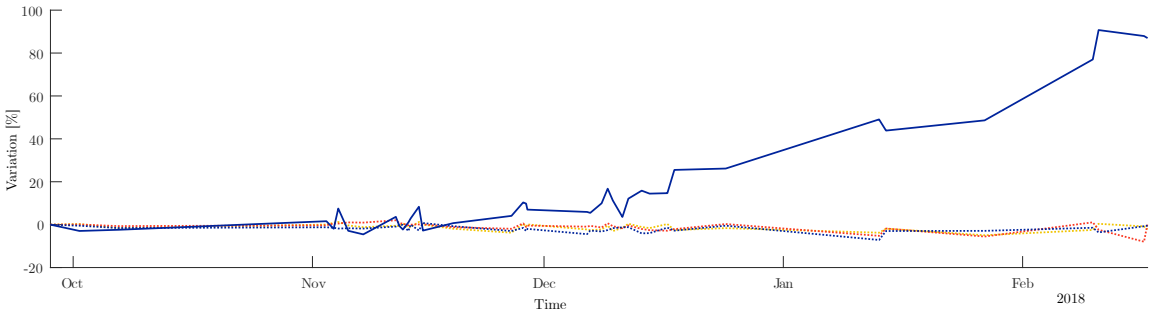

Figure 8. Trend of the EPVA rotor bar passing frequency with sidebands (solid blue line) compared with the raw current spectrum for each phase (red, blue and yellow dotted lines).

The trend in figure 8 elucidates that the increase in rotor deformation steadily developed over a three-month time span. Cross-linking the weather conditions during that period, it became clear that heavy rains had caused muddy water containing many pieces of wood to flood the installation (winter storms). Most likely, thick branches caused the screw to become slightly deformed, subsequently deforming the induction machine's rotor. Corresponding corrective advice was given, enabling the asset owner to act accordingly.

In this case, a very slowly rotating shaft quite far down the drive train was accurately diagnosed. MCSA is one of the few monitoring techniques that performs effectively in both rapidly and slowly rotating applications. As MCSA can determine the rotor's exact position in the stator housing as a quantity that represents physical displacement, no high-frequency content is required to obtain an accurate diagnostic analysis.

5.4 Greasing problem

An industrial steel pickling line, containing several bridle units, is powered by large induction motors. One of these 355 kW two-pole-pair motors is periodically monitored using a handheld vibration measurement device. The high-frequency vibration analysis indicates some deteriorated bearing lubrication (increased noise floor in the envelope spectrum). As the machine is equipped with an online continuous motor current monitoring system, the question arises if this diagnosis can be validated in the electrical data.

An issue with bearing grease quality results in a kind of increased friction, affecting the high-frequency content of an acquired acceleration signal (vibration sensor). At this premature stage, no actual raceway damage has occurred, and the bearing should not yet be replaced. Rather, this early signaling provides the opportunity to proactively correct the root problem (improve lubricant quality) before damage does occur.

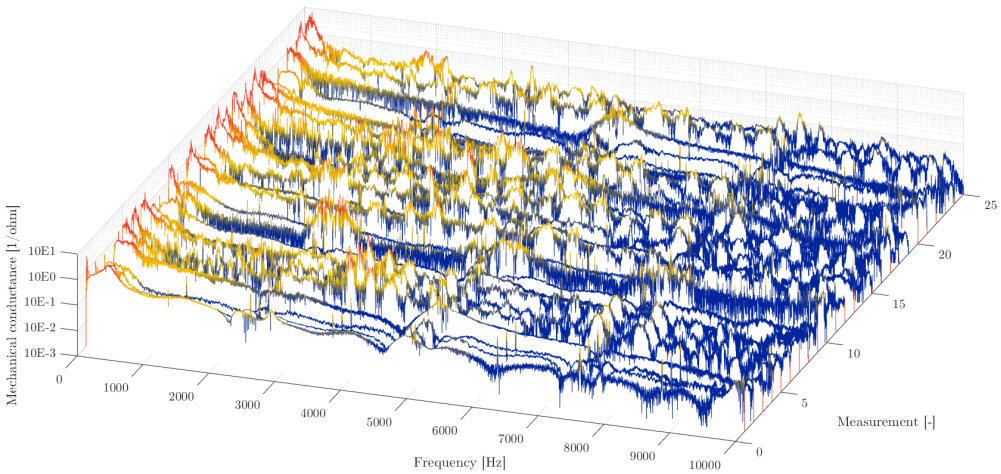

Figure 9. The high-frequency electrical spectrum shows only minor and non-quantified deviations over time (red vs. blue).

Because these early changes occur at such high frequencies, they do not translate into a detectable displacement between the stator and rotor. Consequently, this type of fault is barely visible using MCSA. Figure 9 shows an observable difference in the current signal between sufficient and insufficient greasing, but there is no clear trend over time and the analysis is lacking reproducibility and the theoretical basis to use it with confidence in the diagnostic process. That said, MCSA is still a relatively young technique, and considering its innovative character, it is probable that robust models will eventually be developed for these types of failure modes, moving them from undetectable to detectable.

6 Conclusion

The added value of a condition monitoring system is fully determined by the match between failure mode probability and the propagation of these modes toward the monitoring sensor. MCSA is proven to be one of the more safe, accurate and advanced techniques to detect and diagnose a wide variety of common mechanical and electrical, direct and indirect causes of failure. The combination of advanced signal processing techniques and high-quality interpretation expertise enables trustworthy analysis and targeted corrective maintenance advice.

This paper presented signal processing techniques to expand the scope of MCSA diagnosis from the electric motor to include the transmission and load. By using current and voltage signals to extract the lateral and torsional vibrations in the machine, the motor becomes an extremely sensitive sensor which picks up most drive train operational deviations. Based on the continuing innovative evolution in this field of expertise, we predict that voltage-extended MCSA will eventually become the most effective condition monitoring technique for drive trains powered by electric motors.

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